Probability Distributions and Their Maximum Likelihood Estimators

Let very tiny

$$X_1, X_2, \ldots, X_n$$

be an i.i.d. sample from the distribution under consideration.
The likelihood function is

$$L(\theta) = \prod_{i=1}^{n} f(X_i; \theta),$$

and the maximum likelihood estimator (MLE) is

$$\widehat{\theta}_{\text{MLE}} = \arg\max_{\theta} L(\theta).$$

It is usually convenient to maximize the log-likelihood

$$\ell(\theta) = \log L(\theta).$$

Throughout, let

$$\bar{X} = \frac{1}{n}\sum_{i=1}^{n} X_i, \qquad X_{(1)} = \min_i X_i, \qquad X_{(n)} = \max_i X_i.$$

1 Bernoulli(p)

$$X \sim \text{Bernoulli}(p), \qquad 0 < p < 1.$$

The PMF is

$$P(X = x) = p^x (1-p)^{1-x}, \qquad x \in \{0, 1\}.$$

The likelihood is

$$L(p) = \prod_{i=1}^{n} p^{X_i}(1-p)^{1-X_i}.$$

Thus,

$$L(p) = p^{\sum X_i}(1-p)^{n-\sum X_i}.$$

The log-likelihood is

$$\ell(p) = \left(\sum X_i\right)\log p + \left(n - \sum X_i\right)\log(1-p).$$

Differentiating,

$$\frac{d\ell}{dp} = \frac{\sum X_i}{p} - \frac{n - \sum X_i}{1-p}.$$

Setting this equal to zero,

$$\frac{\sum X_i}{p} = \frac{n - \sum X_i}{1-p}.$$

Therefore,

$$\boxed{\widehat{p} = \frac{1}{n}\sum_{i=1}^{n} X_i = \bar{X}.}$$

2 Binomial(m, p)

Let

$$X \sim \text{Binomial}(m, p),$$

where m is known and $0 < p < 1$.
The PMF is

$$P(X = x) = \binom{m}{x} p^x (1-p)^{m-x}, \qquad x = 0, 1, \ldots, m.$$

The likelihood is

$$L(p) = \prod_{i=1}^{n} \binom{m}{X_i} p^{X_i} (1-p)^{m-X_i}.$$

Ignoring factors that do not depend on p,

$$L(p) \propto p^{\sum X_i} (1-p)^{nm - \sum X_i}.$$

Hence,

$$\ell(p) = \left(\sum X_i\right) \log p + \left(nm - \sum X_i\right) \log(1-p) + C.$$

Differentiating,

$$\frac{d\ell}{dp} = \frac{\sum X_i}{p} - \frac{nm - \sum X_i}{1-p}.$$

Setting the derivative equal to zero gives

$$\boxed{\widehat{p} = \frac{\sum X_i}{nm} = \frac{\bar{X}}{m}.}$$

3 Geometric(p)

We use the convention that X is the number of trials until the first success:

$$X \sim \text{Geometric}(p), \qquad X = 1, 2, \ldots$$

The PMF is

$$P(X = x) = p(1-p)^{x-1}.$$

The likelihood is

$$L(p) = \prod_{i=1}^{n} p(1-p)^{X_i - 1}.$$

Therefore,

$$L(p) = p^n (1-p)^{\sum X_i - n}.$$

The log-likelihood is

$$\ell(p) = n \log p + \left(\sum X_i - n \right) \log(1-p).$$

Differentiating,

$$\frac{d\ell}{dp} = \frac{n}{p} - \frac{\sum X_i - n}{1-p}.$$

Setting this equal to zero,

$$\frac{n}{p} = \frac{\sum X_i - n}{1-p}.$$

Solving,

$$\boxed{\widehat{p} = \frac{n}{\sum X_i} = \frac{1}{\bar{X}}.}$$

Note: If the geometric distribution is instead defined as the number of failures before the first success, the MLE becomes

$$\widehat{p} = \frac{1}{1 + \bar{X}}.$$

4 Poisson(λ)

Let

$$X \sim \text{Poisson}(\lambda), \qquad \lambda > 0.$$

The PMF is

$$P(X = x) = \frac{e^{-\lambda} \lambda^x}{x!}.$$

The likelihood is

$$L(\lambda) = \prod_{i=1}^{n} \frac{e^{-\lambda} \lambda^{X_i}}{X_i!}.$$

Thus,

$$L(\lambda) = e^{-n\lambda}\lambda^{\sum X_i}\prod_{i=1}^{n}\frac{1}{X_i!}.$$

The log-likelihood is

$$\ell(\lambda) = -n\lambda + \left(\sum X_i\right)\log\lambda - \sum\log(X_i!).$$

Differentiating,

$$\frac{d\ell}{d\lambda} = -n + \frac{\sum X_i}{\lambda}.$$

Setting this equal to zero,

$$-n + \frac{\sum X_i}{\lambda} = 0.$$

Hence,

$$\boxed{\widehat{\lambda} = \frac{1}{n}\sum_{i=1}^{n} X_i = \bar{X}.}$$

5 Uniform(a, b)

Let

$$X \sim U(a, b), \qquad a < b.$$

The PDF is

$$f(x; a, b) = \frac{1}{b-a}\mathbf{1}_{\{a\le x\le b\}}.$$

For an i.i.d. sample,

$$L(a, b) = \prod_{i=1}^{n}\frac{1}{b-a}\mathbf{1}_{\{a\le X_i\le b\}}.$$

Therefore,

$$L(a, b) = \frac{1}{(b-a)^n}\mathbf{1}_{\{a\le X_{(1)},\, X_{(n)}\le b\}}.$$

For the likelihood to be nonzero, we require

$$a \le X_{(1)}$$

and

$$b \ge X_{(n)}.$$

Among all such (a, b), we want to minimize

$$b - a,$$

because

$$\frac{1}{(b-a)^n}$$

is maximized when $b-a$ is as small as possible.

The smallest possible interval containing all observations is

$$[X_{(1)}, X_{(n)}].$$

Therefore,

$$\boxed{\widehat{a} = X_{(1)} = \min_i X_i}$$

and

$$\boxed{\widehat{b} = X_{(n)} = \max_i X_i.}$$

Thus,

$$\boxed{(\widehat{a}, \widehat{b}) = (X_{(1)}, X_{(n)}).}$$

Important: This is an example where simply differentiating the log-likelihood with respect to the parameters is not enough, because the support of the distribution depends on the parameters.

6 Normal(μ, σ^2)

Let

$$X \sim N(\mu, \sigma^2), \qquad -\infty < \mu < \infty, \quad \sigma^2 > 0.$$

The PDF is

$$f(x; \mu, \sigma^2) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left[-\frac{(x-\mu)^2}{2\sigma^2}\right].$$

The likelihood is

$$L(\mu, \sigma^2) = (2\pi\sigma^2)^{-n/2} \exp\left[-\frac{1}{2\sigma^2}\sum_{i=1}^{n}(X_i - \mu)^2\right].$$

The log-likelihood is

$$\ell(\mu, \sigma^2) = -\frac{n}{2}\log(2\pi) - \frac{n}{2}\log(\sigma^2) - \frac{1}{2\sigma^2}\sum_{i=1}^{n}(X_i - \mu)^2.$$

MLE of μ

Differentiate with respect to μ:

$$\frac{\partial \ell}{\partial \mu} = \frac{1}{\sigma^2}\sum_{i=1}^{n}(X_i - \mu).$$

Setting this equal to zero,

$$\sum_{i=1}^{n}(X_i - \mu) = 0.$$

Hence,

$$\boxed{\widehat{\mu} = \bar{X}.}$$

MLE of σ^2

Differentiate with respect to σ^2:

$$\frac{\partial \ell}{\partial \sigma^2} = -\frac{n}{2\sigma^2} + \frac{1}{2(\sigma^2)^2}\sum_{i=1}^{n}(X_i - \mu)^2.$$

Setting this equal to zero gives

$$-\frac{n}{2\sigma^2} + \frac{1}{2(\sigma^2)^2}\sum_{i=1}^{n}(X_i - \mu)^2 = 0.$$

Therefore,

$$n\sigma^2 = \sum_{i=1}^{n}(X_i - \mu)^2.$$

Using $\widehat{\mu} = \bar{X}$,

$$\boxed{\widehat{\sigma^2} = \frac{1}{n}\sum_{i=1}^{n}(X_i - \bar{X})^2.}$$

Thus the joint MLE is

$$\boxed{\widehat{\mu} = \bar{X}, \qquad \widehat{\sigma^2} = \frac{1}{n}\sum_{i=1}^{n}(X_i - \bar{X})^2.}$$

Important: The MLE has denominator n, not $n-1$. The estimator

$$S^2 = \frac{1}{n-1}\sum_{i=1}^{n}(X_i - \bar{X})^2$$

is unbiased for σ^2, but it is not the MLE.

7 Exponential(λ)

Let

$$X \sim \text{Exponential}(\lambda), \qquad \lambda > 0,$$

where λ is the **rate** parameter.
The PDF is

$$f(x;\lambda) = \lambda e^{-\lambda x}, \qquad x \geq 0.$$

The likelihood is

$$L(\lambda) = \prod_{i=1}^{n} \lambda e^{-\lambda X_i}.$$

Thus,

$$L(\lambda) = \lambda^n e^{-\lambda \sum X_i}.$$

The log-likelihood is

$$\ell(\lambda) = n \log \lambda - \lambda \sum X_i.$$

Differentiating,

$$\frac{d\ell}{d\lambda} = \frac{n}{\lambda} - \sum X_i.$$

Setting this equal to zero,

$$\frac{n}{\lambda} = \sum X_i.$$

Therefore,

$$\boxed{\widehat{\lambda} = \frac{n}{\sum X_i} = \frac{1}{\bar{X}}.}$$

Parameterization note: If the exponential distribution is parameterized using its mean/scale

$$\theta = \frac{1}{\lambda},$$

then the MLE is

$$\boxed{\widehat{\theta} = \bar{X}.}$$

8 Summary

Distribution	**Parameters**	**MLE**
Bernoulli(p)	p	$\widehat{p} = \bar{X}$
Binomial(m, p)	p, m known	$\widehat{p} = \dfrac{\bar{X}}{m}$
Geometric(p)	p	$\widehat{p} = \dfrac{1}{\bar{X}}$
Poisson(λ)	λ	$\widehat{\lambda} = \bar{X}$
Uniform(a, b)	a, b	$\widehat{a} = X_{(1)}, \quad \widehat{b} = X_{(n)}$
Normal(μ, σ^2)	μ, σ^2	$\widehat{\mu} = \bar{X}, \quad \widehat{\sigma^2} = \dfrac{1}{n}\sum (X_i - \bar{X})^2$
Exponential(λ)	λ (rate)	$\widehat{\lambda} = \dfrac{1}{\bar{X}}$